

SOLAMI: Social Vision-Language-Action Modeling for Immersive Interaction with 3D Autonomous Characters

Jianping Jiang^{*†1}, Weiye Xiao^{*‡1},
 Zhengyu Lin¹, Huaizhong Zhang¹, Tianxiang Ren¹, Yang Gao¹, Zhiqian Lin¹,
 Zhongang Cai^{§1,2}, Lei Yang^{§1}, Ziwei Liu^{§2}
¹SenseTime Research, ²S-Lab, Nanyang Technological University

^{*} Equal Contribution, [†] Project Lead, [‡] Engineering Lead, [§] Corresponding Author

<https://solami-ai.github.io/>

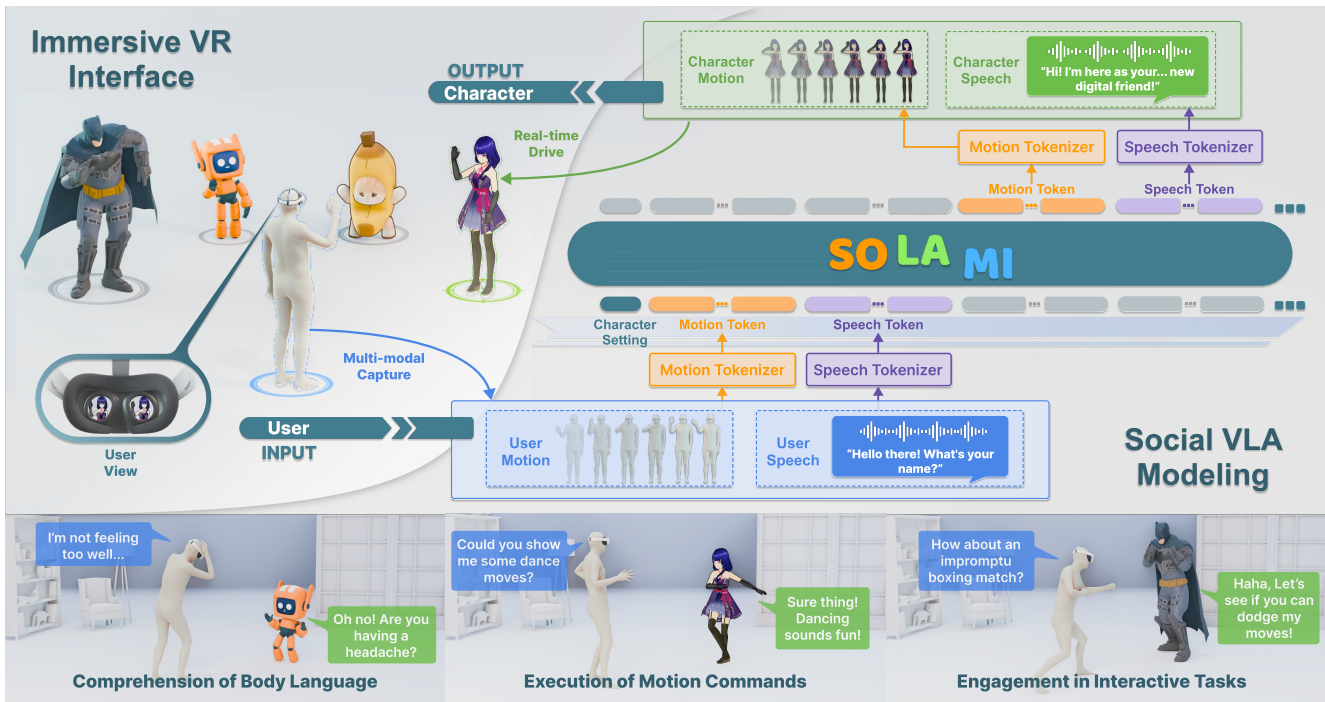


Figure 1. **SOLAMI** enables the user to interact with 3D autonomous characters through speech and body language in an **immersive VR environment** via an end-to-end social vision-language-action model, which is trained on our synthesized multimodal dataset **SynMSI**.

Abstract

Human beings are social animals. How to equip 3D autonomous characters with similar social intelligence that can perceive, understand and interact with humans remains an open yet fundamental problem. In this paper, we introduce **SOLAMI**, the first end-to-end **Social vision-Language-Action (VLA) Modeling** framework for **Immersive interaction** with 3D autonomous characters. Specifically, **SOLAMI** builds 3D autonomous characters from three aspects: **1) Social VLA Architecture:** We propose a unified social VLA framework to generate multi-modal response (speech and motion) based on the user's

multimodal input to drive the character for social interaction. **2) Interactive Multimodal Data:** We present **SynMSI**, a **synthetic multimodal social interaction** dataset generated by an automatic pipeline using only existing motion datasets to address the issue of data scarcity. **3) Immersive VR Interface:** We develop a VR interface that enables users to immersively interact with these characters driven by various architectures. Extensive quantitative experiments and user study demonstrate that our framework leads to more precise and natural character responses (in both speech and motion) that align with user expectations with lower latency.

1. Introduction

Have you ever imagined having an immersive face-to-face conversation with a character you deeply admire? Not merely through speech dialogue, but through an interaction where you can observe its subtle facial expressions, natural body language, and even fleeting emotional changes. Psychology research [15, 25, 46, 49] shows that in social interactions, the greater the level of immersion, the better the human experience. However, current character agents [1, 3, 47] are still limited to text or voice interactions. This limitation prompts us to build 3D autonomous characters with richer modalities.

Developing an autonomous 3D character requires effectively modeling its behavior system, which involves two major challenges: 1) The 3D character needs to accurately observe and understand the information conveyed by the user, and respond appropriately based on the context and its character setting through speech, body motion, and facial expression, *etc.* This goes beyond previous singular human-related tasks, such as motion generation [62], motion understanding [27], and audio-to-motion [6]. 2) Data for multimodal interactions between users and 3D characters is extremely scarce due to the prohibitive cost of the comprehensive setup.

Previous work [13] is primarily based on the LLM-Agent framework, using text to link various sub-modules (such as motion captioning and text-to-motion). While this approach performs well in high-level tasks like planning and memory, it tends to fall short in tasks such as understanding user behaviors and providing timely body motion responses. This limitation arises because using text as the intermediary between modules conveys high-level information but often omits subtle nuances. And the sub-modules (motion captioning, speech recognition, *etc.*) in the complex engineering framework incur substantial latency that undermines the timeliness of a natural communication [50].

Interestingly, research on robotics shows the similar conclusion. The LLM-Agent framework can handle planning tasks [26], but for low-level manipulation tasks, end-to-end Vision-Language-Action (VLA) models built upon LLMs show superior performance [11, 24, 29, 67]. We argue that digital avatars are essentially robots with virtual humanoid embodiment. Therefore, building a VLA model for social interactions with users is a promising direction.

In this paper, we implement an end-to-end social VLA model, **SOLAMI**. Our model, built upon a decoder-only LLM backbone, processes the inputs of user speech and motion into discrete representations, and generate the responsive speech and motion tokens, which are then decoded to the character’s speech and motion. This modeling approach can effectively learn character behavior patterns across motion and speech modalities and offer low latency.

Although there are numerous datasets related to human

social behaviors [13, 34, 40, 56], comprehensive multimodal interaction datasets remain scarce. Thus, we introduce a data synthesis method that utilizes existing text-motion datasets to automatically construct multimodal interaction data at a low cost. Leveraging our extensively curated topics (5.3 K), uniformly processed motion database (46 K), and iterative script refinement pipeline, we develop **SynMSI**, a dataset containing 6.3 K multi-turn multimodal conversation items. To evaluate the effectiveness of our method, we developed a VR interface where users can immersively interact with various 3D characters. Quantitative experimental results and user study analysis show that our approach produces more precise and natural social interaction experience with lower latency.

To summarize, we contribute **1) A new VLA architecture** to model the character’s behavior system for immersive social interaction; **2) A dedicated synthesizing pipeline** that automatically generates large-scale multimodal interactive dataset, SynMSI; **3) An immerse VR interface** for users to interact with various characters through speech and motion.

2. Related Work

2.1. Human Motion & LLM

Existing role-play agents primarily rely on text [47], speech [1, 3, 41], or video [7] as interactive media, but building 3D autonomous characters means modeling 3D embodied behavior, especially body language. Unlike single-purpose human motion tasks [6, 12, 22, 34, 48, 56, 62], we expect this 3D character not only to comprehend user speech and body language but also to respond according to its profile setting. Large language models (LLMs) [41, 51], with their remarkable emergent abilities [54], provide promising solutions for this direction. One approach [18, 28, 64, 65] tries to integrate LLMs and human motion in an end-to-end fashion, enabling a single model to perform multiple motion-related tasks. However, the goal of these methods is not to generate responsive motion based on the input motion according to the character setting. Another approach [13, 35] utilizes the LLM as a versatile brain center that controls various sub-modules (e.g., motion understanding, text-to-motion generation) with text or code instructions. However, this modular approach inherently introduces information loss and engineering-related time latency. Therefore, how to create feasible autonomous 3D characters for immersive interaction remains an open challenge.

2.2. Embodied Intelligence

Building 3D autonomous characters means creating virtual humanoid agents with embodied intelligence. In the field of embodied intelligence, researchers have found the high-

level abilities (planning, memory, etc.) of LLM-Agents across various environments, including factories [16, 26], 2D sandbox settings [42], and 3D gaming spaces [53]. For tasks requiring low-level skills, such as manipulation, end-to-end VLA models [11, 24, 29, 67] have shown considerable potential. Despite existing efforts [19, 20, 22] in ego-centric human-related tasks, the capabilities of VLA models in social interactive tasks with humans have not been fully explored.

3. Social Vision-Language-Action Modeling

3.1. Architecture

Our framework is an end-to-end social VLA model that takes the user’s speech and motion as input and generates the character’s responsive speech and motion as output. In this process, speech and motion are added as new languages to the LLM text vocabulary. First, the user’s speech and motion are converted into discrete motion tokens and speech tokens via a motion tokenizer and a speech tokenizer, respectively. A decoder-only LLM backbone then predicts the character’s output motion and speech tokens based on the user’s input tokens and the character setting. The generated tokens are subsequently decoded into corresponding motion and speech by their respective decoders.

Motion Representation. To take advantage of SMPL-X’s [43] compatibility with industry animation workflow, we directly model human poses as SMPL-X joint rotations instead of keypoint positions [28, 38] to facilitate animation of characters down the stream.

Motion Tokenizer. Our motion tokenizer employs a Vector Quantized Variational Autoencoders (VQ-VAE) structure as [28, 64]. It learns discrete representations of motion, enabling the LLM to understand the text-motion connection. We design separate VQVAEs (Q^b, Q^h, Q^t) for the body motion $\mathbf{m}^b$, hand motion $\mathbf{m}^h$, and inter-character relative transform (rotation and translation) $\mathbf{m}_t$ for higher reconstruction accuracy [38]. The quantization process for motion $\mathbf{m}^u$ is formulated as

$$\hat{\mathbf{m}}_t^u = Q^u(\mathbf{m}_t^u) = \arg \min_{z_i \in \mathbb{Z}_u} \|\mathbf{m}_t^u - z_i\|_2, \quad (1)$$

where $\mathbb{Z}_u$ is the codebook of motion part $u \in \{b, h, t\}$, and $\hat{\mathbf{m}}_t^u$ is the corresponding motion tokens. The VQ-VAEs of body and hand, Q^b and Q^h , apply 1D convolutions on motion features along the temporal dimension to get L_M sequential tokens $\hat{M}^b = [\hat{m}_1^b, \hat{m}_2^b, \dots, \hat{m}_{L_M}^b]$ and $\hat{M}^h = [\hat{m}_1^h, \hat{m}_2^h, \dots, \hat{m}_{L_M}^h]$, and the VQVAE Q^t uses MLPs to get the transform token $\hat{m}^t$ of a sequence.

Speech Tokenizer. Research [9, 61] on speech discretization mainly utilizes the RVQ-VAE structure [59]. In this work, we utilize the SpeechTokenizer [63] that disentangles semantic and acoustic information within speech S . This

allows us to use the semantic tokens $\hat{S} = [\hat{s}_1^s, \hat{s}_2^s, \dots, \hat{s}_{L_S}^s]$ from the first layer as input to the LLM, reducing inference costs (L_S is the sequence length of semantic tokens). Simultaneously, a short sample of the character’s voice (4 to 6 seconds) can be used as a prompt to achieve instance voice cloning when decoding through SoundStorm [10].

Multi-modal Multi-round Interaction. User-character interaction is formulated as a multi-round conversation fashion of common LLMs [41, 51]. When the user sends speech and motion to SOLAMI, the model auto-regressively generates speech and motion responses based on previous dialogue contents and the character setting. To facilitate training, we use special tokens to mark the start and end of each modality sequence as [28, 30]. The interaction process can be formulated as follow:

Interaction Template

```
Input:
System Prompt: <Character_Placeholder>
User: <M_Placeholder><S_Placeholder>
Character: <M_Placeholder><S_Placeholder>
...
User: <M_Placeholder><S_Placeholder>
Output:
Character: <M_Placeholder><S_Placeholder>
```

where $\langle \text{Character_Placeholder} \rangle$ is placeholder for character’s text description, $\langle \text{M_Placeholder} \rangle$ for motion token sequences, and $\langle \text{S_Placeholder} \rangle$ for speech token sequences.

3.2. Training

The training of SOLAMI adopts a three-stage strategy.

Stage 1: Tokenizer Training. The training approach for the motion tokenizer uses the fashion of [28]. The train loss is

$$\mathcal{L}_m = \lambda_r \mathcal{L}_r + \lambda_e \mathcal{L}_e + \lambda_c \mathcal{L}_c + \lambda_v \mathcal{L}_v, \quad (2)$$

where $\mathcal{L}_r$ means reconstruction loss, $\mathcal{L}_e$ embedding loss, $\mathcal{L}_c$ commitment loss, $\mathcal{L}_v$ velocity loss, and λ_* are manually adjusted weights. For the speech tokenizer, we use the pre-trained checkpoint from AnyGPT [60]. We freeze the tokenizers’ weights after this stage.

Stage 2: Multi-task Pre-training for Modality Alignment. As shown in Fig. 2, the second stage is multi-task pre-training, achieving modality alignment between motion and text, as well as between speech and text. It is necessary because motion data is scarce, and direct training on multimodal interaction data results in sub-optimal models, as demonstrated in subsequent ablation studies. For motion and text alignment, we use 46 K motion-text pairs for text-to-motion generation and motion captioning tasks, and 11 K interactive motion pairs

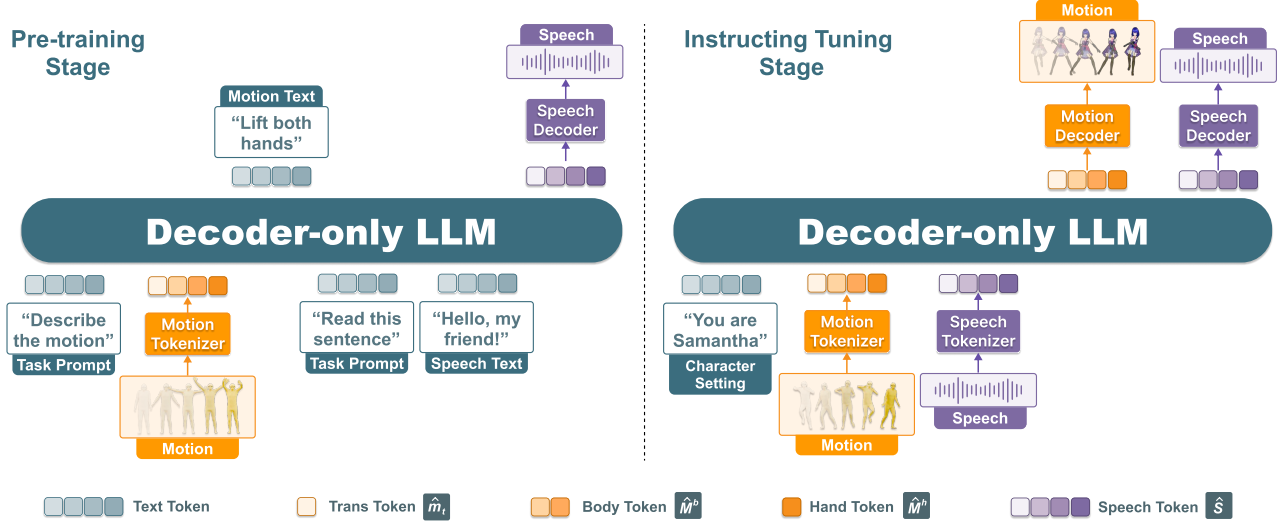


Figure 2. Training pipeline of SOLAMI. We train SOLAMI through a three-stage process. In the pre-training stage, we train the model with motion-text and speech-text related tasks to align the speech and motion modalities with language. During the instruction tuning stage, we train the model with social multimodal multi-round interaction data, enabling it to generate multimodal responses that align with the character settings and the context of the topic.

for two-person motion generation. To align the speech and text, we train the model with 410 K speech-text pairs for text-to-speech and automatic speech recognition tasks, and 100 K speech dialogue pairs for speech-to-speech generation. The tasks are formulated as “User: $\langle \text{Task_Placeholder} \rangle \langle \text{Input_Modality_Placeholder} \rangle$; Character: $\langle \text{Output_Modality_Placeholder} \rangle$ ”. To balance the scale disparity between the motion and speech data, we sampled them at a 4:6 ratio during training.

Stage 3: Instruction Tuning for Multi-turn Conversation. In the third stage, we perform instruction tuning by applying supervised fine-tuning with multimodal interaction data, enabling the model to handle long-sequence, multi-turn dialogues, as shown in Fig. 2. We utilize 5.7 K multimodal conversation items for supervised fine-tuning. Each conversation item is fed to the model in the format as the Interaction Template in Sec. 3.1. We experiment with two approaches: full-parameter fine-tuning and LoRA fine-tuning [23]. We supervised only the character’s response to teach the model how to react to the user’s behavior. Thus we train the model by maximizing the log-likelihood of the next-token prediction and the train loss is:

$$\begin{aligned} \mathcal{L}_{\text{IT}} = & - \sum_{r=1}^R \sum_{i=1}^{L_M^r} \log p_{\Theta}(\hat{m}_i^r | \hat{m}_{i-1}^r, \dots, \hat{m}_1^r, \hat{S}_{<r}, \hat{M}_{<r}) \\ & - \sum_{r=1}^R \sum_{i=1}^{L_S^r} \log p_{\Theta}(\hat{s}_i^r | \hat{s}_{i-1}^r, \dots, \hat{s}_1^r, \hat{S}_{<r}, \hat{M}_{\leq r}), \end{aligned} \quad (3)$$

where Θ is the network parameter of the LLM backbone or LoRA weights, R is the conversation round, $\hat{S}_r$ and $\hat{M}_r$

are the r -th round speech and motion token sequences with lengths L_M^r and L_S^r , respectively.

4. SynMSI Dataset

Social interaction between users and virtual characters is inherently unique, which makes collecting such multimodal interaction data particularly challenging. Currently, available public datasets [21, 40, 56] are incomplete for our task. To address this issue, we propose a data synthesis pipeline that leverages existing motion-text datasets, text-based role-play models, and speech synthesis methods and generates a large-scale multimodal dialogue dataset, **SynMSI**.

4.1. Motion Data

We collect motion-text data for two purposes: first, to achieve alignment between motion and text during pre-training, and second, to generate multimodal data for instruction tuning. Since our work focuses on modeling social interactions, we select existing datasets that contain rich social actions: HumanML3D [21] with 24 K motion-text pairs, Inter-X [56] with 20 K motion-text pairs and 10 K two-person motion pairs, and DLP-MoCap [13] with 2 K motion-text pairs. Since the Inter-X [56] dataset contains only text descriptions of two-person interactive motion without descriptions of individual motion, we used GPT-4o [41] to decompose the two-person action descriptions into single-person motion-text pairs. Additionally, we used GPT-4o [41] to synthesize comprehensive text descriptions for each motion clip by consolidating multiple possible descriptions, thereby providing more detailed textual annotations that preserve motion details.

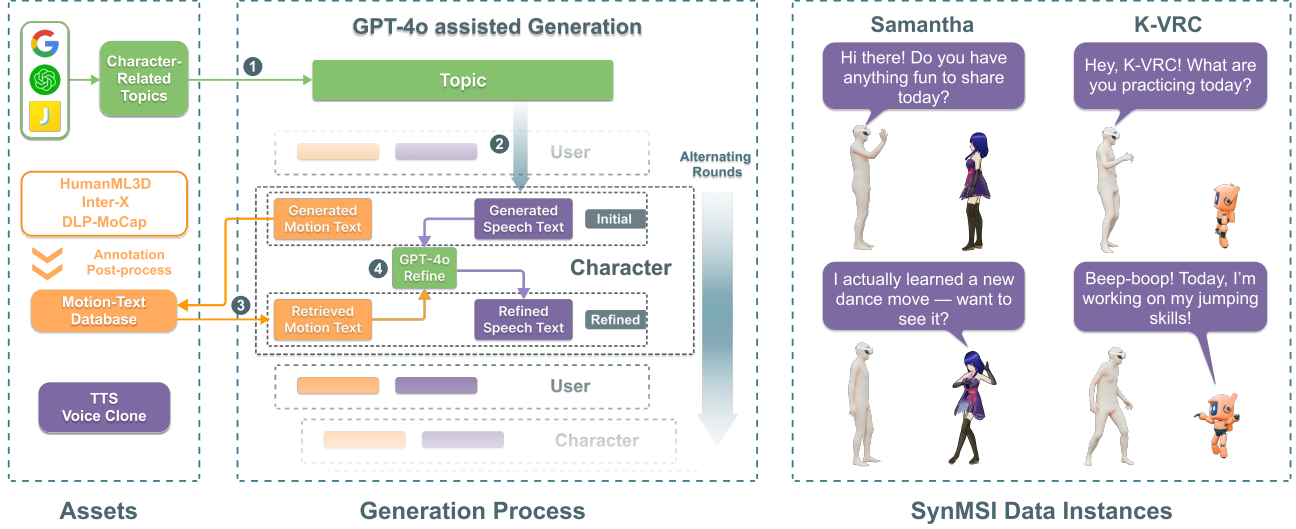


Figure 3. SynMSI dataset generation. Our synthesizing pipeline consists of 4 steps. Based on numerous character-relevant topics and state-of-the-art LLMs [41], we generate text scripts for multimodal dialogues. Using a large-scale motion database [13, 21, 56], we retrieve the most appropriate motions and refine the speech scripts accordingly. Finally, we employ TTS/voice cloning [14] to generate character-specific speech. This approach enables us to create multimodal interaction data of various characters using only existing motion datasets.

4.2. Speech Data

We use speech-text data for speech-text alignment in the pre-training stage. Speech datasets involve Common-Voice [8] (150 K speech-text pairs in our experiments), AnyInstruct [60] (200 K speech-text pairs and 100 K speech-to-speech items), and synthetic speech data (60 K speech-text pairs) by text-to-speech approaches (Azure TTS and XTTS_v2 [14]).

4.3. Multimodal Data Synthesizing

Previous data synthesis methods [32, 37, 66] usually use the general abilities of advanced LLMs [41] and text-annotated multimodal data to generate synthetic data. However, generating social multimodal interaction data has not yet been achieved. This challenging task requires high-quality expression of body language, voice consistency that matches the characters, and suitable dialogue content.

As shown in Fig. 3, our synthesizing pipeline includes 4 steps. (1) First, we collect 5.3 K character-related and daily topics from internet platforms (Google Trends [2], Zhihu [5], Jike) and brainstorm of GPT-4o [41] to improve the diversity of the dialogue contents. (2) Based on the topic, character setting, and previous round of scripts, we use GPT-4o [41] to generate textual descriptions (motion, speech, expression *etc.*) for the next round of the dialogue. (3) Then we utilize the text embedding [39] of the motion description to retrieve the most relevant motions from our meticulously curated motion-text database. (4) Moreover, we refine the generated speech text with the retrieved motions to ensure that speech and motion are well-coordinated.

The motion database with detailed text annotations and the refinement process can alleviate the misalignment between the real motion and speech in the LLM-Agent method [13]. By iteratively repeating this process, we can generate multi-round dialogue contents across many characters, where the motions are sourced from the motion database, and the speeches are synthesized using TTS/voice cloning (Azure TTS and XTTS_v2 [14]) to maintain consistency with the character’s voice style. We finally obtained 6.3 K multimodal dialogue items.

5. VR Interface

To demonstrate our method directly, we developed a VR interface with an Oculus Quest 3 frontend and a backend service, as shown in Fig. 4. The frontend enables immersive interaction between users and 3D autonomous characters, while the backend, powered by 2 H800 GPUs, supports the computation of various baselines in our experiments. During usage, the VR headset captures the user’s speech and body motion, and sends them to the backend computation nodes. For motion capture, we use the Quest’s full-body tracking system [55] to obtain pose parameters, which are then retargeted onto an SMPL-X model [43]. The computation nodes generate the body motion parameters and speech responses of the character based on the multimodal input. Then we apply an audio-to-face method, UniTalker [17], to generate the character’s facial animation parameters. The facial and body parameters are jointly retargeted onto a 3D character model [58], completing one cycle of social interaction. For a natural user experience, we employ preset idle

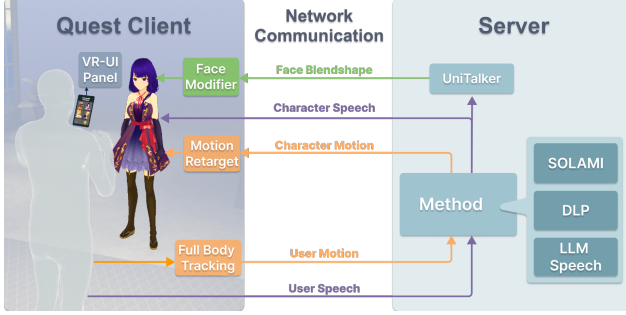


Figure 4. VR interface architecture. Our VR project consists of a Quest 3 client and a server. The Quest client captures and transmits user body motion and speech to the server. The server then generates character’s speech, body motion, and face blendshape parameters based on the selected methods. The response is then sent back to the Quest client to drive the character.

motions on the character when the method is LLM+Speech or the character is waiting for the user’s input.

3D Character Assets. Our 3D character portfolio covers a diverse range of entities, including AI assistant avatars, famous cinematic roles, internet memes, and real-world celebrity personas. These 3D models are sourced from open-source repositories under *CC Attribution-NonCommercial-ShareAlike License* as well as our manual creation using VRoid Studio [4]. We subsequently employ facial rigging, skinning, bone chain simulation, retargeting, and texture and material creation in Unity Engine processes to yield functional 3D character assets.

6. Experiments

6.1. Experimental Settings

In our experiment, we selected the AnyGPT-base model [60] (based on LLaMA2-7B [51]) as the backbone for SOLAMI, because it is an open-source model available at our experimental time that supports end-to-end speech processing. During the pre-training stage, we utilize 32 V100 GPUs to train the model for 3 K steps (batch size 256, learning rate 4e-5). For instruction tuning, we train the SOLAMI for 800 steps using 16 V100 GPUs (batch size 48, learning rate 2e-5). For LoRA fine-tuning [23], we set the rank as 8 and alpha as 16. We split the synthesized multimodal data into training and test sets with a 9:1 ratio. We use DeepSpeed [45] to accelerate the training. During testing, we evaluate each round of the character’s response.

Baselines. To validate the performance improvement in social interaction brought by incorporating 3D modalities (such as body motion), we compared SOLAMI with the *LLM+Speech* and the *AnyGPT (fine-tune)* approach. For the *LLM+Speech* framework, the user’s speech is first transcribed into text using ASR techniques, which is then fed to a LLM to generate the character’s response in text, and sub-

sequently converted into speech using TTS. For fairness, we use LLaMA2-7B-Chat [51] as the LLM backbone, Whisper large-v3 [44] for ASR, and XTTS_v2 [14] for voice cloning. For the *AnyGPT (fine-tune)* framework, we use the speech data of SynMSI to train the AnyGPT-base model [60] with the same parameter settings. To compare the effectiveness of the LLM-Agent architecture with the social VLA framework, we used *DLP* [13] as a baseline method. In *DLP* framework, the user’s speech and body motion are separately processed as text descriptions by ASR and motion captioning modules. Based on the input text descriptions, LLM generates the character’s text instructions of speech and motion, which are transferred into speech and body motion by TTS and motion generation module. The speech component of *DLP* follows the *LLM+Speech* method. Considering that MoMat-MoGen module in *DLP* is too slow for user interaction (over 5 seconds latency), we use MotionGPT [28] for motion captioning and motion generation. To ensure a fair comparison, we used the same motion data as the pre-training stage of SOLAMI to train MotionGPT. Additionally, we conducted ablation experiments on the effect of the pre-training stage, marked as (*w/o pretrain*). For the ablation study of the motion tokenizer, please refer to the supplementary materials. We use vLLM [31] to accelerate the LLM backbones for low latency interaction.

6.2. Quantitative Evaluation

We conducted quantitative evaluation for our method and all the baselines mentioned in Sec. 6.1.

Evaluation Metrics. For motion, we evaluate the model responses using metrics including FID, diversity, PAMPIPE (mm), and angle error [22]. Following Duolando [48], we obtain FID and diversity using motion features from AIST++ [33]. For speech, we use VC similarity [57] to evaluate the voice similarity with the character. To evaluate the content quality of speech, we first use Whisper-large-v3 [44] to transcribe the speech into text. Then following [47, 52], we employ GPT-4o [41] as the judge to assess *Context Relevance* and *Character Consistency* on a Likert scale ranging from 1 to 5. *Context Relevance* indicates whether the speech content aligns with the topic and context of the conversation, while *Character Consistency* assesses whether the content adheres to the character settings. For inference latency (seconds), we deploy all the models on 2 H800 GPUs with vLLM [31] framework and asynchronous mechanisms to improve performance while maintaining fairness.

Quantitative Results. The quantitative results in Tab. 1 demonstrate that, using the same foundation model (Llama2) [51] as the backbone, our method, SOLAMI (full params), significantly outperforms other methods in terms of motion quality and inference latency.

Motion Quality. SOLAMI demonstrates superior perfor-

Table 1. **Quantitative results of baselines and SOLAMI.** ‘↑’(‘↓’) indicates that the values are better if the metrics are larger (smaller). We run all the evaluations 5 times and report the average metric. The best results are in bold and the second best results are underlined.

Methods	Motion Metrics				Speech Metrics			Inference Latency ↓
	FID↓	Diversity↑	PA-MPJPE↓	Angle Error↓	VC Similarity↑	Context Relevance↑	Character Consistency↑	
SynMSI Dataset	-	9.136	-	-	-	4.888	4.893	-
LLM+Speech (Llama2) [51]	-	-	-	-	0.818	3.527	3.859	3.157
AnyGPT (fine-tune) [60]	-	-	-	-	0.819	3.502	3.803	2.588
DLP (MotionGPT) [13]	<u>4.254</u>	8.259	165.053	0.495	0.812	<u>3.577</u>	3.785	5.518
SOLAMI (w/o pretrain)	5.052	<u>8.558</u>	<u>159.709</u>	<u>0.387</u>	<u>0.820</u>	3.541	3.461	2.657
SOLAMI (LoRA)	15.729	8.145	167.149	0.400	0.770	3.251	3.423	2.710
SOLAMI (full params)	3.443	8.853	151.500	0.360	0.824	3.634	<u>3.824</u>	<u>2.639</u>

mance compared with the DLP method [13] across multiple motion metrics. This is because the social VLA model achieves comprehensive modality alignment among speech, motion, and language through training on our high-fidelity character-specific multimodal data. Our model can precisely perceive the user’s physical motions and linguistic cues, enabling semantically rich interactive motions in response. This ability contrasts with the LLM-Agent architectures of the DLP method, which exhibits limitations in conveying multimodal nuances through text-only intermediary representation.

Speech Quality. Our method demonstrates the capability to synthesize a voice tone that matches the character with a higher Voice Cloning (VC) Similarity score. Our model also shows better performance on the context relevance of the dialogue than other methods. Because the inclusion of the motion modality enables the model to perceive the user’s body language, while the LLM+Speech or AnyGPT(fine-tune) method lacks this capability. In terms of character consistency, our model achieves secondary performance metrics. We suspect this may be due to the incorporation of motion and speech modalities, which potentially affects the character-related knowledge embedded within the original LLM. The performance degradation is similar to the observations in [16, 36].

Inference Latency. Our end-to-end approach is significantly superior to the modular pipeline approaches (LLM+Speech or DLP). Because the end-to-end VLA model naturally aligns with the process of real-time human communication. Theoretically, if we could collect data on real-time interactions between humans and characters, our method could achieve full-duplex streaming interaction.

Ablation Study. As shown in Tab. 1, the pre-training stage of SOLAMI leads to better performance in both motion and speech quality. We believe that the pre-training stage, which aligns motion, speech, and language, facilitates the model’s ability to learn the multimodal dialogue skill during the instruction tuning stage. Instruction tuning using LoRA [23] shows weaker results compared to full parameter fine-tuning. We think that the gap between the data distribution of pre-training tasks and the multimodal instruction tuning task is substantial, and LoRA fine-tuning alone is insufficient for the model to learn the strong multimodal

Table 2. Questionnaire settings of our user study.

Dimension	Questions
Motion Coherence	Does the motion match the character’s setting?
	Does the motion align well with speech?
Motion Interaction	Does the character follow motion instructions correctly?
	Does the character understand user’s motion?
Speech Consistency	Does the speech match the character’s setting?
	Is the speech relevant to the current topic?
Overall Experience	How would you rate the overall experience?

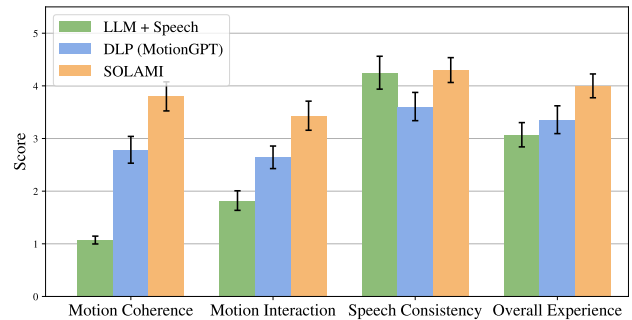


Figure 5. Results of the user study with 95% confidence.

dialog ability.

6.3. VR Interface User Study

Quantitative evaluation of a single modality alone cannot fully compare the 3D autonomous characters built on different frameworks. To address this, we conducted user study with a VR interface. As shown in Fig. 4, we employ LLM+Speech, DLP (MotionGPT), and SOLAMI as method options of the server backend and the same VR frontend across different methods. This implies that the variable in the experiment is the driving method in the server. Users are asked to engage in more than five rounds of dialogue with the VR character before completing our questionnaire.

Evaluation Metrics. The indicators and corresponding questions for our questionnaire are shown in Tab. 2. *Motion Coherence* evaluates whether the character’s motion aligns with the character setting and dialogue content. *Motion Interaction* assesses whether the character can understand the semantics of the body motion and effectively interact with the user. *Speech Coherence* examines whether the generated speech aligns with the context and character settings.

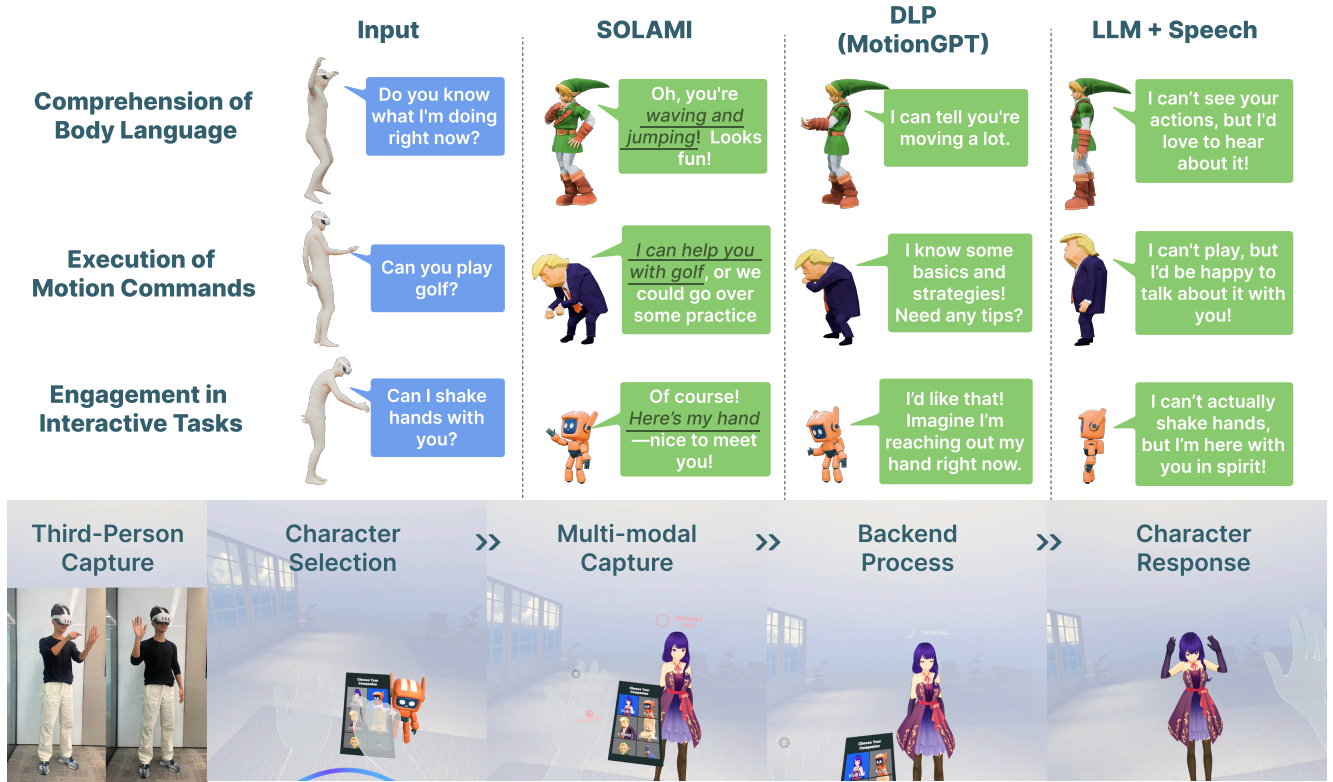


Figure 6. Qualitative results of SOLAMI and baselines, and the user workflow for VR experience. Our social VLA model, trained in an end-to-end strategy on SynMSI dataset, can accurately perceive the semantic information embedded within users’ speech and motion input, and subsequently generate natural and coherent responses.

And the *Overall Experience* measures the user’s satisfaction with the overall experience. Each question is rated on a 1 to 5 Likert scale, with higher scores indicating greater satisfaction. The score for each dimension is calculated as the average score of its corresponding questions. We ultimately collected 60 survey responses, with participants from various gender and age groups.

Results. As shown in the Fig. 5, our method achieved the highest user satisfaction across all dimensions. SOLAMI demonstrates superior performance over the DLP method across all metrics, validating the effectiveness of an end-to-end social VLA model in character behavior modeling. While the DLP method shows lower speech consistency compared to the LLM+Speech method, it excels in motion-related metrics and achieves higher overall satisfaction, indicating that effective body language can enhance user experience despite speech quality limitations.

To provide a more intuitive demonstration of our model’s capabilities, we replay the user study experiments and render them, as shown in the Fig. 6. The results indicate that SOLAMI demonstrates excellent capabilities in body language understanding, motion command execution, and body interaction. For better understanding, we also present the workflow from a first-person view during actual usage.

We encourage readers to explore our supplementary materials and video for a more detailed overview of the model experiments, data generation pipeline, VR interface construction process, and comprehensive experimental results.

7. Conclusion

In this paper, we propose SOLAMI, an approach for building 3D autonomous characters. This approach includes three key components: 1) *Architecture*: A novel social VLA modeling framework enabling multimodal social interaction; 2) *Multimodal Data Synthesizing*: A pipeline for automatically generating multimodal interaction data from existing incomplete datasets; 3) *VR Interface*: A VR engineering framework that facilitates immersive interactions between users and various characters. Together, these modules contribute to an enhanced user interaction experience.

Acknowledgement

This study is supported by the Ministry of Education, Singapore, under its MOE AcRF Tier 2 (MOET2EP20221-0012, MOE-T2EP20223-0002), and under the RIE2020 Industry Alignment Fund – Industry Collaboration Projects (IAF-ICP) Funding Initiative, as well as cash and in-kind contribution from the industry partner(s).

References

- [1] Character.ai. <https://character.ai>. 2
- [2] Google trends. <https://trends.google.com/trends>. 5
- [3] Talkie ai. <https://www.talkie-ai.com>. 2
- [4] Vroid studio. <https://vroid.com/en/studio>. 6
- [5] Zhihu. <https://www.zhihu.com>. 5
- [6] Simon Alexanderson, Rajmund Nagy, Jonas Beskow, and Gustav Eje Henter. Listen, denoise, action! audio-driven motion synthesis with diffusion models. *ACM TOG*, 2023. 2
- [7] Tenglong Ao. Body of her: A preliminary study on end-to-end humanoid agent. *arXiv preprint arXiv:2408.02879*, 2024. 2
- [8] Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis M. Tyers, and Gregor Weber. Common voice: A massively-multilingual speech corpus. In *LREC*, 2020. 5
- [9] Zálán Borsos, Raphaël Marinier, Damien Vincent, Eugene Kharitonov, Olivier Pietquin, Matthew Sharifi, Dominik Roblek, Olivier Teboul, David Grangier, Marco Tagliasacchi, and Neil Zeghidour. Audioldm: A language modeling approach to audio generation. *TASLP*, 2023. 3
- [10] Zálán Borsos, Matt Sharifi, Damien Vincent, Eugene Kharitonov, Neil Zeghidour, and Marco Tagliasacchi. Soundstorm: Efficient parallel audio generation. *arXiv preprint arXiv:2305.09636*, 2023. 3
- [11] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil J. Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl Pertsch, Jornell Quiambao, Kanishka Rao, Michael S. Ryoo, Grecia Salazar, Pannag R. Sanketi, Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong T. Tran, Vincent Vanhoucke, Steve Vega, Quan Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. RT-1: robotics transformer for real-world control at scale. In *RSS*, 2023. 2, 3
- [12] Zhongang Cai, Wanqi Yin, Ailing Zeng, Chen Wei, Qingping Sun, Wang Yanjun, Hui En Pang, Haiyi Mei, Mingyuan Zhang, Lei Zhang, Chen Change Loy, Lei Yang, and Ziwei Liu. Smpler-x: Scaling up expressive human pose and shape estimation. In *NeurIPS*, 2023. 2
- [13] Zhongang Cai, Jianping Jiang, Zhongfei Qing, Xinying Guo, Mingyuan Zhang, Zhengyu Lin, Haiyi Mei, Chen Wei, Ruisi Wang, Wanqi Yin, Liang Pan, Xiangyu Fan, Han Du, Peng Gao, Zhitao Yang, Yang Gao, Jiaqi Li, Tianxiang Ren, Yukun Wei, Xiaogang Wang, Chen Change Loy, Lei Yang, and Ziwei Liu. Digital life project: Autonomous 3d characters with social intelligence. In *CVPR*, 2024. 2, 4, 5, 6, 7
- [14] Edresson Casanova, Kelly Davis, Eren Gölge, Gökem Gökner, Iulian Gulea, Logan Hart, Aya Aljafari, Joshua Meyer, Reuben Morais, Samuel Olayemi, et al. Xtts: a massively multilingual zero-shot text-to-speech model. *arXiv preprint arXiv:2406.04904*, 2024. 5, 6
- [15] James J Cummings and Jeremy N Bailenson. How immersive is enough? A meta-analysis of the effect of immersive technology on user presence. *Media psychology*, 2016. 2
- [16] Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc Toussaint, Klaus Greff, Andy Zeng, Igor Mordatch, and Pete Florence. Palm-e: An embodied multimodal language model. In *ICML*, 2023. 3, 7
- [17] Xiangyu Fan, Jiaqi Li, Zhiqian Lin, Weiye Xiao, and Lei Yang. Unitalker: Scaling up audio-driven 3d facial animation through A unified model. 2024. 5
- [18] Yao Feng, Jing Lin, Sai Kumar Dwivedi, Yu Sun, Priyanka Patel, and Michael J. Black. Chatpose: Chatting about 3d human pose. In *CVPR*, 2024. 2
- [19] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, Miguel Martín, Tushar Nagarajan, Ilija Radosavovic, Santhosh Kumar Ramakrishnan, Fiona Ryan, Jayant Sharma, Michael Wray, Mengmeng Xu, Eric Zhongcong Xu, Chen Zhao, Siddhant Bansal, Dhruv Batra, Vincent Cartillier, Sean Crane, Tien Do, Morrie Doulaty, Akshay Erapalli, Christoph Feichtenhofer, Adriano Fragomeni, Qichen Fu, Abrahm Gebreselasie, Cristina González, James Hillis, Xuhua Huang, Yifei Huang, Wenqi Jia, Weslie Khoo, Jáchym Kolár, Satwik Kotur, Anurag Kumar, Federico Landini, Chao Li, Yanghao Li, Zhenqiang Li, Karttikeya Mangalam, Raghava Modhugu, Jonathan Munro, Tullie Murrell, Takumi Nishiyasu, Will Price, Paola Ruiz Puentes, Merey Ramazanov, Leda Sari, Kiran Somasundaram, Audrey Southerland, Yusuke Sugano, Ruijie Tao, Minh Vo, Yuchen Wang, Xindi Wu, Takuma Yagi, Ziwei Zhao, Yunyi Zhu, Pablo Arbeláez, David Crandall, Dima Damen, Giovanni Maria Farinella, Christian Fuegen, Bernard Ghanem, Vamsi Krishna Ithapu, C. V. Jawahar, Hanbyul Joo, Kris Kitani, Haizhou Li, Richard A. Newcombe, Aude Oliva, Hyun Soo Park, James M. Rehg, Yoichi Sato, Jianbo Shi, Mike Zheng Shou, Antonio Torralba, Lorenzo Torresani, Mingfei Yan, and Jitendra Malik. Ego4d: Around the world in 3, 000 hours of egocentric video. In *CVPR*, 2022. 3
- [20] Kristen Grauman, Andrew Westbury, Lorenzo Torresani, Kris Kitani, Jitendra Malik, Triantafyllos Afouras, Kumar Ashutosh, Vijay Baiyya, Siddhant Bansal, Bikram Boote, Eugene Byrne, Zachary Chavis, Joya Chen, Feng Cheng, Fu-Jen Chu, Sean Crane, Avijit Dasgupta, Jing Dong, María Escobar, Cristhian Forigua, Abrahm Gebreselasie, Sanjay Haresh, Jing Huang, Md Mohaiminul Islam, Suyog Dutt Jain, Rawal Khiradkar, Devansh Kukreja, Kevin J. Liang, Jia-Wei Liu, Sagnik Majumder, Yongsan Mao, Miguel Martín, Effrosyni Mavroudi, Tushar Nagarajan, Francesco Ragusa, Santhosh Kumar Ramakrishnan, Luigi Seminara, Arjun Somayazulu, Yale Song, Shan Su, Zihui Xue, Ed-

- ward Zhang, Jinxu Zhang, Angela Castillo, Changan Chen, Xinzhu Fu, Ryosuke Furuta, Cristina González, Prince Gupta, Jiabo Hu, Yifei Huang, Yiming Huang, Weslie Khoo, Anush Kumar, Robert Kuo, Sach Lakhavani, Miao Liu, Mi Luo, Zhengyi Luo, Brighid Meredith, Austin Miller, Oluwatumininu Oguntola, Xiaqing Pan, Penny Peng, Shraman Pramanick, Merey Ramazanova, Fiona Ryan, Wei Shan, Kiran Somasundaram, Chenan Song, Audrey Southerland, Masatoshi Tateno, Huiyu Wang, Yuchen Wang, Takuma Yagi, Mingfei Yan, Xitong Yang, Zecheng Yu, Shengxin Cindy Zha, Chen Zhao, Ziwei Zhao, Zhifan Zhu, Jeff Zhuo, Pablo Arbeláez, Gedas Bertasius, Dima Damen, Jakob Engel, Giovanni Maria Farinella, Antonino Furnari, Bernard Ghanem, Judy Hoffman, C. V. Jawahar, Richard A. Newcombe, Hyun Soo Park, James M. Rehg, Yoichi Sato, Manolis Savva, Jianbo Shi, Mike Zheng Shout, and Michael Wray. Ego-exo4d: Understanding skilled human activity from first- and third-person perspectives. In *CVPR*, 2024. 3
- [21] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating diverse and natural 3d human motions from text. In *CVPR*, 2022. 4, 5
- [22] Fangzhou Hong, Vladimir Guзов, Hyo Jin Kim, Yuting Ye, Richard Newcombe, Ziwei Liu, and Lingni Ma. EgoIm: Multi-modal language model of egocentric motions. *arXiv preprint arXiv:2409.18127*, 2024. 2, 3, 6
- [23] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *ICLR*, 2022. 4, 6, 7
- [24] Jiangyong Huang, Silong Yong, Xiaojian Ma, Xiongkun Linghu, Puhao Li, Yan Wang, Qing Li, Song-Chun Zhu, Baoxiong Jia, and Siyuan Huang. An embodied generalist agent in 3d world. In *ICML*, 2024. 2, 3
- [25] Sarah Hudson, Sheila Matson-Barkat, Nico Pallamin, and Guillaume Jegou. With or without you? Interaction and immersion in a virtual reality experience. *Journal of business research*, 2019. 2
- [26] Brian Ichter, Anthony Brohan, Yevgen Chebotar, Chelsea Finn, Karol Hausman, Alexander Herzog, Daniel Ho, Julian Ibarz, Alex Irpan, Eric Jang, Ryan Julian, Dmitry Kalashnikov, Sergey Levine, Yao Lu, Carolina Parada, Kanishk Rao, Pierre Sermanet, Alexander Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Mengyuan Yan, Noah Brown, Michael Ahn, Omar Cortes, Nicolas Sievers, Clayton Tan, Sichun Xu, Diego Reyes, Jarek Rettinghouse, Jor-nell Quiambao, Peter Pastor, Linda Luu, Kuang-Huei Lee, Yuheng Kuang, Sally Jesmonth, Nikhil J. Joshi, Kyle Jeffrey, Rosario Jauregui Ruano, Jasmine Hsu, Keerthana Gopalakrishnan, Byron David, Andy Zeng, and Chuyuan Kelly Fu. Do as I can, not as I say: Grounding language in robotic affordances. In *CoRL*, 2022. 2, 3
- [27] Biao Jiang, Xin Chen, Wen Liu, Jingyi Yu, Gang Yu, and Tao Chen. Motiongpt: Human motion as a foreign language. In *NeurIPS*, 2023. 2
- [28] Biao Jiang, Xin Chen, Wen Liu, Jingyi Yu, Gang Yu, and Tao Chen. Motiongpt: Human motion as a foreign language. In *NeurIPS*, 2023. 2, 3, 6
- [29] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Paul Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024. 2, 3
- [30] Dan Kondratyuk, Lijun Yu, Xiuye Gu, José Lezama, Jonathan Huang, Grant Schindler, Rachel Hornung, Vigh-nesh Birodkar, Jimmy Yan, Ming-Chang Chiu, Krishna Somandepalli, Hassan Akbari, Yair Alon, Yong Cheng, Joshua V. Dillon, Agrim Gupta, Meera Hahn, Anja Hauth, David Hendon, Alonso Martinez, David Minnen, Mikhail Sirotenko, Kihyuk Sohn, Xuan Yang, Hartwig Adam, Ming-Hsuan Yang, Irfan Essa, Huisheng Wang, David A. Ross, Bryan Seybold, and Lu Jiang. Videopoet: A large language model for zero-shot video generation. In *ICML*, 2024. 3
- [31] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles*, 2023. 6
- [32] KunChang Li, Yinan He, Yi Wang, Yizhuo Li, Wenhai Wang, Ping Luo, Yali Wang, Limin Wang, and Yu Qiao. Videochat: Chat-centric video understanding. *arXiv preprint arXiv:2305.06355*, 2023. 5
- [33] Ruilong Li, Shan Yang, David A. Ross, and Angjoo Kanazawa. AI choreographer: Music conditioned 3d dance generation with AIST++. In *ICCV*, 2021. 6
- [34] Han Liang, Wenqian Zhang, Wenxuan Li, Jingyi Yu, and Lan Xu. Intergen: Diffusion-based multi-human motion generation under complex interactions. *IJCV*, 2024. 2
- [35] Jing Lin, Yao Feng, Weiyang Liu, and Michael J Black. Chathuman: Language-driven 3d human understanding with retrieval-augmented tool reasoning. *arXiv preprint arXiv:2405.04533*, 2024. 2
- [36] Ji Lin, Hongxu Yin, Wei Ping, Pavlo Molchanov, Mohammad Shoeybi, and Song Han. VILA: on pre-training for visual language models. In *CVPR*, 2024. 7
- [37] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, 2023. 5
- [38] Shunlin Lu, Ling-Hao Chen, Ailing Zeng, Jing Lin, Ruimao Zhang, Lei Zhang, and Heung-Yeung Shum. Humantomato: Text-aligned whole-body motion generation. In *ICML*, 2024. 3
- [39] Arvind Neelakantan, Tao Xu, Raul Puri, Alec Radford, Jesse Michael Han, Jerry Tworek, Qiming Yuan, Nikolas Tezak, Jong Wook Kim, Chris Hallacy, et al. Text and code embeddings by contrastive pre-training. *arXiv preprint arXiv:2201.10005*, 2022. 5
- [40] Evonne Ng, Javier Romero, Timur M. Bagautdinov, Shaojie Bai, Trevor Darrell, Angjoo Kanazawa, and Alexander Richard. From audio to photoreal embodiment: Synthesizing humans in conversations. In *CVPR*, 2024. 2, 4
- [41] OpenAI. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024. 2, 3, 4, 5, 6

- [42] Joon Sung Park, Joseph C. O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *UIST*, 2023. 3
- [43] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. Expressive body capture: 3d hands, face, and body from a single image. In *CVPR*, 2019. 3, 5
- [44] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *ICML*, 2023. 6
- [45] Jeff Rasley, Samyam Rajbhandari, Olatunji Ruwase, and Yuxiong He. Deepspeed: System optimizations enable training deep learning models with over 100 billion parameters. In *KDD*, 2020. 6
- [46] Giuseppe Riva, Fabrizia Mantovani, Claret S. Capideville, Alessandra Preziosa, Francesca Morganti, Daniela Villani, Andrea Gaggioli, Cristina Botella, and Mariano Alc  niz Raya. Affective interactions using virtual reality: The link between presence and emotions. *Cyberpsychology Behav. Soc. Netw.*, 2007. 2
- [47] Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. Character-llm: A trainable agent for role-playing. In *EMNLP*, 2023. 2, 6
- [48] Li Siyao, Tianpei Gu, Zhitao Yang, Zhengyu Lin, Ziwei Liu, Henghui Ding, Lei Yang, and Chen Change Loy. Duolando: Follower GPT with off-policy reinforcement learning for dance accompaniment. In *ICLR*, 2024. 2, 6
- [49] Mel Slater, Daniel P  rez Marcos, Henrik Ehrsson, and Maria V Sanchez-Vives. Inducing illusory ownership of a virtual body. *Frontiers in neuroscience*, 2009. 2
- [50] Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. Cognitive architectures for language agents. 2024. 2
- [51] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton-Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aur  lien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023. 2, 3, 6, 7
- [52] Quan Tu, Shilong Fan, Zihang Tian, Tianhao Shen, Shuo Shang, Xin Gao, and Rui Yan. Charactereval: A chinese benchmark for role-playing conversational agent evaluation. In *ACL*, 2024. 6
- [53] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *TMLR*, 2024. 3
- [54] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent abilities of large language models. *TMLR*, 2022. 2
- [55] Alexander W. Winkler, Jungdam Won, and Yuting Ye. Questsim: Human motion tracking from sparse sensors with simulated avatars. In *SIGGRAPH Asia*, 2022. 5
- [56] Liang Xu, Xintao Lv, Yichao Yan, Xin Jin, Shuwen Wu, Congsheng Xu, Yifan Liu, Yizhou Zhou, Fengyun Rao, Xingdong Sheng, Yunhui Liu, Wenjun Zeng, and Xiaokang Yang. Inter-x: Towards versatile human-human interaction analysis. In *CVPR*, 2024. 2, 4, 5
- [57] Dongchao Yang, Jinchuan Tian, Xu Tan, Rongjie Huang, Songxiang Liu, Xuankai Chang, Jiatong Shi, Sheng Zhao, Jiang Bian, Xixin Wu, Zhou Zhao, and Helen Meng. Uni-audio: An audio foundation model toward universal audio generation. *arXiv preprint arXiv:2310.00704*, 2023. 6
- [58] Zhitao Yang, Zhongang Cai, Haiyi Mei, Shuai Liu, Zhaoxi Chen, Weiye Xiao, Yukun Wei, Zhongfei Qing, Chen Wei, Bo Dai, Wayne Wu, Chen Qian, Dahua Lin, Ziwei Liu, and Lei Yang. Synbody: Synthetic dataset with layered human models for 3d human perception and modeling. In *ICCV*, 2023. 5
- [59] Neil Zeghidour, Alejandro Luebs, Ahmed Omran, Jan Skoglund, and Marco Tagliasacchi. Soundstream: An end-to-end neural audio codec. *TASLP*, 2022. 3
- [60] Jun Zhan, Junqi Dai, Jiasheng Ye, Yunhua Zhou, Dong Zhang, Zhigeng Liu, Xin Zhang, Ruibin Yuan, Ge Zhang, Linyang Li, Hang Yan, Jie Fu, Tao Gui, Tianxiang Sun, Yungang Jiang, and Xipeng Qiu. Anygpt: Unified multimodal LLM with discrete sequence modeling. In *ACL*, 2024. 3, 5, 6, 7
- [61] Dong Zhang, Shimin Li, Xin Zhang, Jun Zhan, Pengyu Wang, Yaqian Zhou, and Xipeng Qiu. Speechgpt: Empowering large language models with intrinsic cross-modal conversational abilities. In *EMNLP*, 2023. 3
- [62] Mingyuan Zhang, Zhongang Cai, Liang Pan, Fangzhou Hong, Xinying Guo, Lei Yang, and Ziwei Liu. Motiondiffuse: Text-driven human motion generation with diffusion model. *IEEE TPAMI*, 2024. 2
- [63] Xin Zhang, Dong Zhang, Shimin Li, Yaqian Zhou, and Xipeng Qiu. Speechokeizer: Unified speech tokenizer for speech large language models. *arXiv preprint arXiv:2308.16692*, 2023. 3
- [64] Yaqi Zhang, Di Huang, Bin Liu, Shixiang Tang, Yan Lu, Lu Chen, Lei Bai, Qi Chu, Nenghai Yu, and Wanli Ouyang. Motiongpt: Finetuned llms are general-purpose motion generators. In *AAAI*, 2024. 2, 3

- [65] Zixiang Zhou, Yu Wan, and Baoyuan Wang. Avatargpt: All-in-one framework for motion understanding, planning, generation and beyond. In *CVPR*, 2024. [2](#)
- [66] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models. In *ICLR*, 2024. [5](#)
- [67] Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong T. Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: vision-language-action models transfer web knowledge to robotic control. In *CoRL*, 2023. [2](#), [3](#)